# Configuration and secrets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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TEACHER GUIDE · 1 OF 9

## Prepare the purpose, prerequisites and materials

> Prepare Configuration and secrets: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of **configuration** and **secret**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Typed contracts and schemas

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of typed contracts and schemas

Reference example, not the only acceptable wording: Teacher explanation: Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

### Requirements and service boundaries

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of requirements and service boundaries

Reference example, not the only acceptable wording: Teacher explanation: A service boundary states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Two 45-minute sessions. **Configuration** supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the **mechanism** and terminology of **configuration** and **secret**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

### Prerequisite answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

### End goal for the pictorial case

Explain **configuration** and **secret**s using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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# Explain and model the mechanism

> Model Configuration and secrets using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

**Configuration** supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

## Explain Configuration and secrets · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

## Explain Configuration and secrets · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

| Term | Meaning |
| --- | --- |
| Secret | A value whose disclosure can grant privileged access |
| Configuration | Declared settings controlling operation |

| Term | Meaning |
| --- | --- |
| Redaction | Removing sensitive values from displayed or recorded output |

### Configuration and secrets — evidence trace

**INPUT AND SCOPE**

A startup script prints every environment variable while debugging a missing data version.

**CHECK OR CALCULATION**

The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.

**CONCLUSION AND LIMIT**

It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

## Explain Configuration and secrets · 3

Guided: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A startup script prints every environment variable while debugging a missing data version.

It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

## Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Configuration and secrets

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

### The facts we will use

> **Supplied facts**
>
> maximum_words=80 is public **configuration**. A separate protected store holds a

placeholder credential.

AI ACADEMY / AI-6/4

## Configuration and secrets

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Load setting

Read the reviewed maximum_words value of 80.



### 2. Load credential

The runtime obtains the token through its protected path.



### 3. Log safely

Record startup success without printing the token.

**CHANGE ONE THING**

Change maximum_words to 60.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Load setting | Read the reviewed maximum_words value of 80. |
| 2. Load credential | The runtime obtains the token through its protected path. |
| 3. Log safely | Record startup success without printing the token. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

**Configuration** controls behaviour. A credential grants access and needs different storage and handling.

### What this example does not establish

#### Keep the stated limit

Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

### Change one thing in the pictured case

#### Change one thing

Change: Change maximum_words to 60.

#### Predicted result

**Prediction**: The configured answer limit becomes 60; the private token stays unchanged.

#### Why the result follows

Reason: An ordinary length setting does not require replacing the credential.

### Ask while showing the picture

#### Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Configuration and secrets, reduce support deliberately and check individual reasoning before discussing answers.

| Time or stage | Teacher action |
| --- | --- |
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

### Reduce help deliberately

#### Completion practice

Completion task: A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

### Reasoning to listen for

Teacher check: Teacher explanation: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

#### Individual reasoning check

Use the worked case for **Configuration** and **secret**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

#### Pictorial practice question

Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

**Reasoned teacher answer**

Read the reviewed maximum_words value of 80. The runtime obtains the token through its protected path. Record startup success without printing the token.

Changed case: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Changed-case reasoning: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the investigation and diagnose misconceptions

> Run the Configuration and secrets investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Configuration and secrets questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Guided and exit questions

Guided: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### W1 Explain the mechanism

#### Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

#### Teacher reasoning · W1

Teacher explanation: **Configuration** supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

### W2 Trace the worked case

#### Workbook question · W2

A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

#### Teacher reasoning · W2

Teacher explanation: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

### W3 Audit the limit

#### Workbook question · W3

A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

#### Teacher reasoning · W3

Teacher explanation: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

### W4 Apply a new case independently

#### Workbook question · W4

Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

#### Teacher reasoning · W4

Teacher explanation: The server/client boundary leaked the **secret**. Remove value disclosure; record a non-sensitive error ID and missing/invalid key name, and follow the actual **secret**-response procedure.

### W5 Design a revealing follow-up

#### Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

#### Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A classroom mock is not a production credential-management design; no real **secret**s are needed here.

The new case should test this particular **mechanism** rather than merely restate the original result.

### Independent assessment and fresh retry

#### Teacher key · Independent assessment and fresh retry

W4: The server/client boundary leaked the **secret**. Remove value disclosure; record a non-sensitive error ID and missing/invalid key name, and follow the actual **secret**-response procedure.

Require intermediate reasoning and the stated scope. Fresh retry: A debug log stores an access token to explain invalid **configuration**.

Expected reasoning: Redact the token; log the validation category and trace ID rather than credential contents.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: The server/client boundary leaked the **secret**. Remove value disclosure; record a non-sensitive error ID and missing/invalid key name, and follow the actual **secret**-response procedure.

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect the project, support and next lesson

> Connect Configuration and secrets to Grounded Learning Service, adapt support and plan a later changed-case check.

Add a labelled **configuration** and **secret**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **configuration** and **secret**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Authentication and authorisation. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### Plan the delayed return

Suggested return points in this level: ai-6/5, ai-6/7, ai-6/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Configuration and secrets lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Explain the **mechanism** and terminology of **configuration** and **secret**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

### Supported practice answer

Reasoned practice answer: Teacher explanation: It risks exposing **secret**s unrelated to the error. Report the missing key name without values and validate only required fields.

Check every learner before discussion: Use the worked case for **Configuration** and **secret**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Change maximum_words to 60. Require a **prediction** and a reason before comparing. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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## Visual concept journey

> Show one Configuration and secrets graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Configuration and secrets: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Supplied facts: maximum_words=80 is public **configuration**.

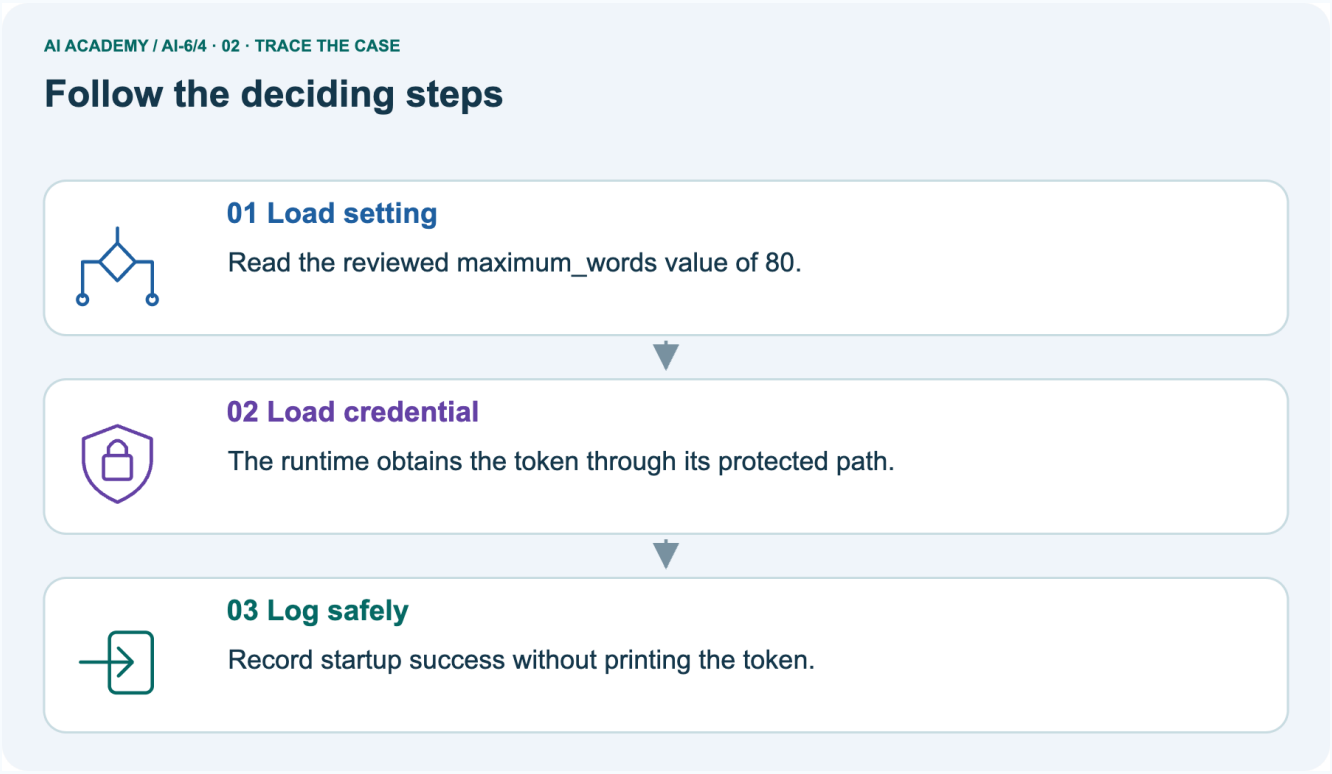
A separate protected store holds a placeholder credential.

> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Load setting: Read the reviewed maximum_words value of 80. → Load credential: The runtime obtains the token through its protected path. → Log safely: Record startup success without printing the token.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Configuration and secrets — evidence trace

AI ACADEMY / AI-6/4 · 03 · INSPECT THE LESSON MODEL

## Configuration and secrets — evidence trace

| Input and scope | Check or calculation | Conclusion and limit |
| --- | --- | --- |
| A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret. | It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

**Configuration** and **secret**s — **evidence** trace.

A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**.

Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**. | It risks exposing **secret**s unrelated to the error.

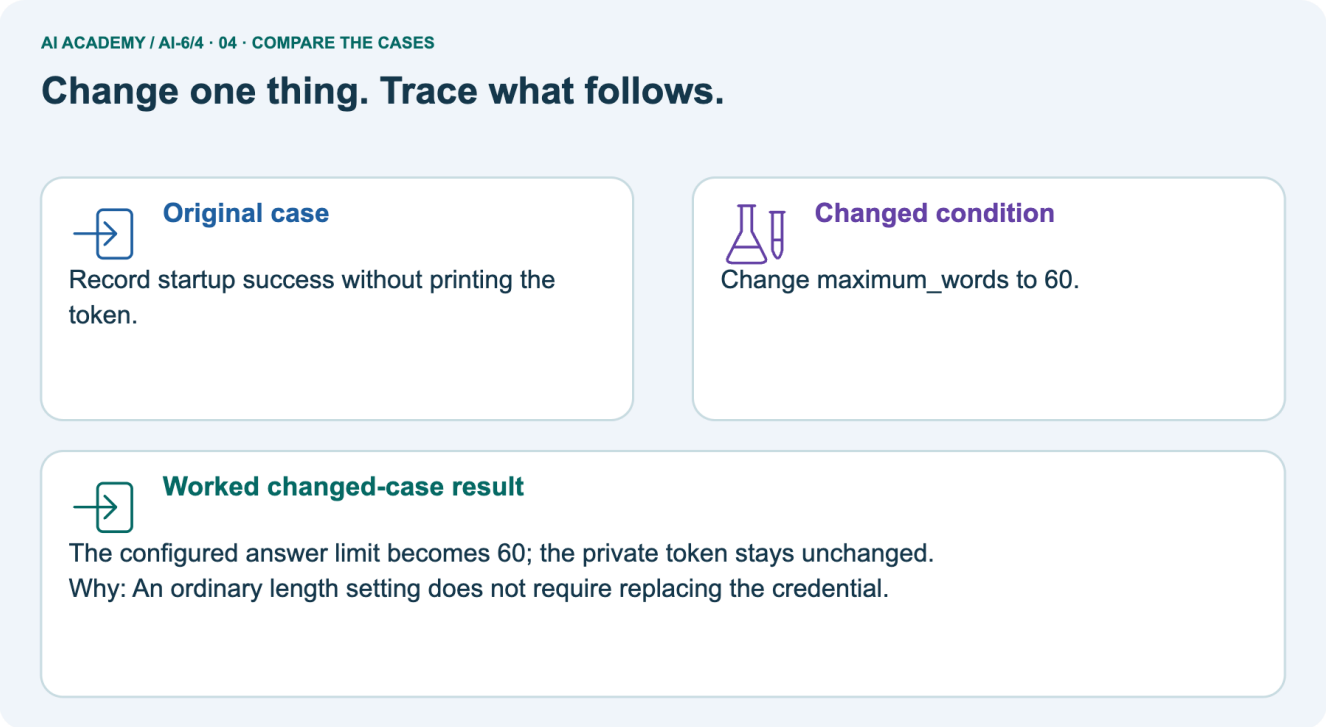
Report the missing key name without values and validate only required fields.

> Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record startup success without printing the token. Changed condition: Change maximum_words to 60. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

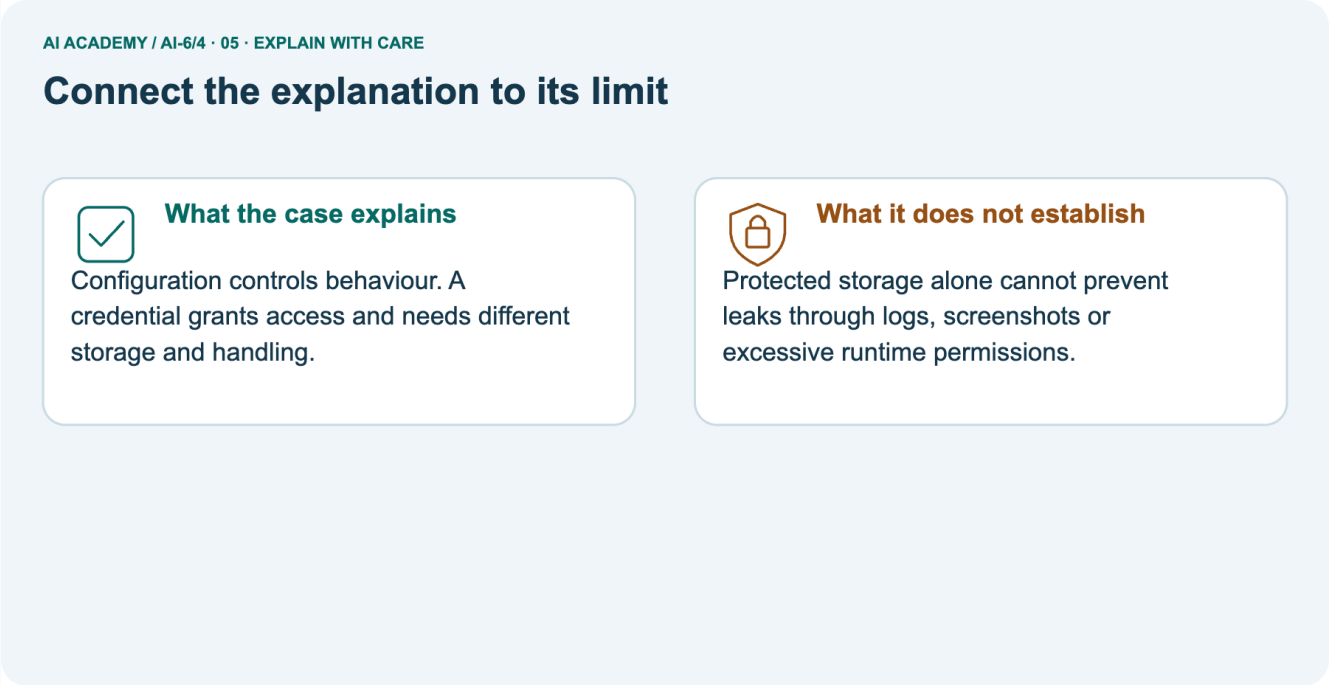
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: An ordinary length setting does not require replacing the credential.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

**Configuration** controls behaviour. A credential grants access and needs different storage and handling. Limit: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

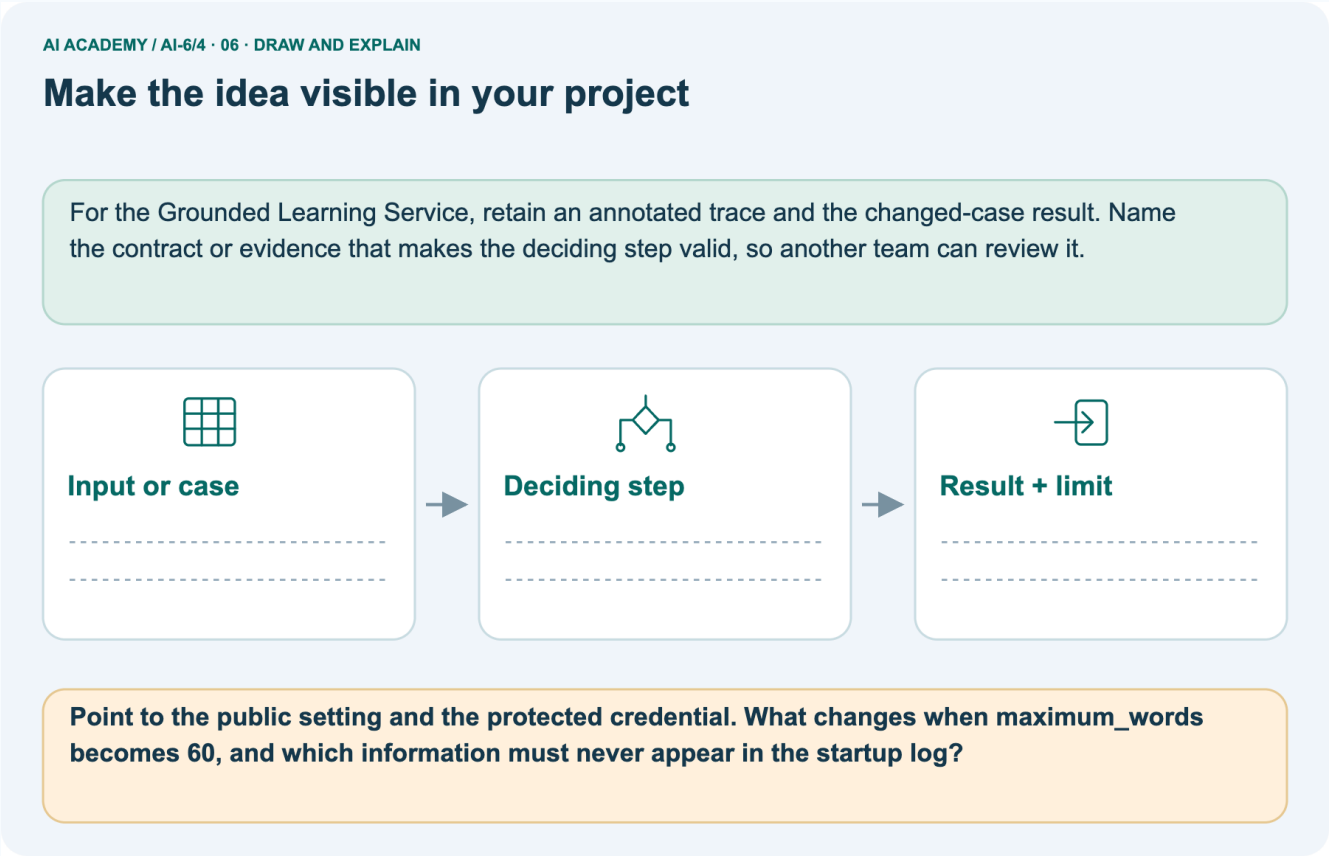
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Read the reviewed maximum_words value of 80. The runtime obtains the token through its protected path. Record startup success without printing the token.

Changed case: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.